

Protection and Pricing Power

in Sequential Venture Financing

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September 23, 2026

Outline

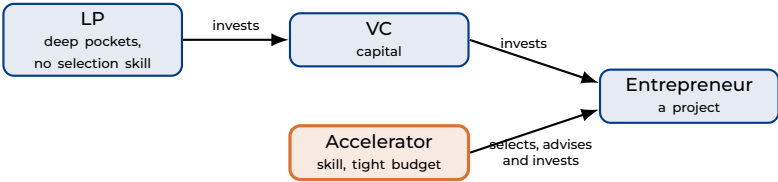
- 1 Introduction
- 2 Literature Review
- 3 Contribution
- 4 Model
- 5 Game 3: the pre-seed investor's own clause
- 6 Estimation
- 7 Takeaways

Who does what in venture capital

The classic chain



With an accelerator



Accelerators contracts

The fixed slice is standardised across accelerators. The contingent tranche is not, and it is where the MFN appears.

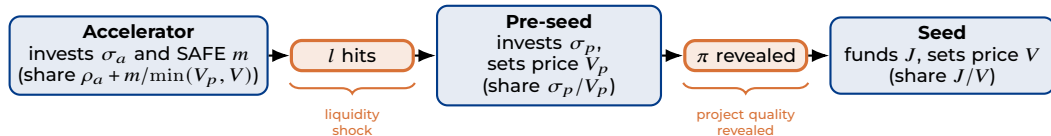
	Fixed amount, fixed share	Contingent tranche
Y Combinator	\$125K for 7%	\$375K uncapped MFN safe, since 2022
Techstars	\$20K for 6% (5% in 2025)	\$100K note capped at \$3M, then MFN in 2025
500 Global	\$150K for 6%	—
Entrepreneur First	\$125K for 8%	\$125K uncapped MFN safe
Antler	£125K for 8.5%	£85K convertible note

Entrepreneur First and Antler: London and UK programmes.

Research question

What does an MFN clause held by the first investor do to the rounds that follow — to the stake each party ends up with, to the amounts they put in, and to the seed probability of projects?

Most Favored Nation explained



- **Sequential contracting.** (McAfee and Schwartz, 1994; Hart and Tirole, 1990)
 - MFN as a commitment remedy — studied for the grantor, here for a third party.
- **Staging and convertibles.** (Gompers, 1995; Kaplan and Strömberg, 2003; Cornelli and Yosha, 2003; Schmidt, 2003)
 - Devices against moral hazard. Here the conversion *basis* is the object.
- **Accelerators.** (Gonzalez-Uribe and Leatherbee, 2017; Hallen et al., 2020)
 - The contract that finances them has not been modelled.
- **Interim finance.** (Admati and Pfleiderer, 1994; Hellmann and Thiele, 2015)
 - Here a claim held by an *earlier* investor, converting on the interim price.

This paper

The accelerator is the only winner of the MFN introduction. The pre-seed investor strictly loses. The probability of seed success and the seed investor's gain are both ambiguous.

Striking findings

- **Accelerator.** Holds $\rho_a^0 + m/\min(V, V_p)$ in place of ρ_a^0 and earns more at every shock, most where its tranche converts lowest.
- **Pre-seed investor.** Earns less at every shock, even where it pre-funds and ends up holding more of the company than in the benchmark.
- **Seed investor.** Same amount, same stake, same payoff where the pre-seed investor advances the shock and no more. Where it pre-funds, the seed investor funds less at a lower threshold, and its payoff has no sign.
- **Projects.** No sign. Fewer where the accelerator's cash covers the shock on its own: the tranche converts at the round and raises the bar by the factor $1 + m/I$. Weakly more where a pre-seed round does occur, and exactly as many where that investor advances the shock and no more.

This paper and YC's claims

Announcing the switch to uncapped MFN in 2022, YC's then-president argued that the new deal terms would:

- **remove the pressure to fundraise on unfavourable terms.**
 - Yes. Fewer firms need a pre-seed round at all, and those that do are priced no lower.
- **make a successful seed round more likely.**
 - Only through the cash. The clause itself takes m off what the pre-seed investor will advance, so companies in that band lose their investor and the round with it; and a company that needs no pre-seed round has nobody to absorb the clause, so its threshold rises. What the cash adds is what decides the sign.

The agents

The project

The round must cover $I \in (0, 1)$: the investment, and the cost c_E the entrepreneur must be left to run the project. Outcome in $\{0, 1\}$ with success probability π , density f . A liquidity shock l hits after the accelerator has invested.

Assumptions

- **(A1)** The average project is worth financing: $\mathbb{E}(\pi - l) - I > 0$ (and f log-concave).
- **(B)** Let β the pre-seed investor's budget: $\beta \leq I$.
- **(C)** The entrepreneur runs the project only if his share covers his cost: $\rho_E \pi \geq c_E$. It is the one participation constraint that survives.

Why (B) is important

Without it the investor funds the round and takes the entire residual equity.

Game 1 (Benchmark)

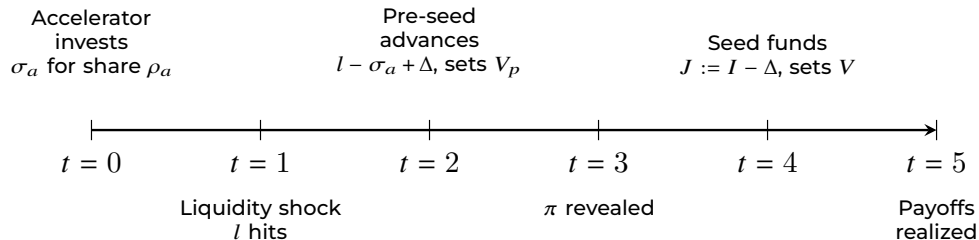


Figure: Game 1 (Benchmark)

Game 1 (Benchmark)

The seed investor maximizes its utility:

$$\max_V \int_{\pi_{\min}}^1 \left(\frac{J}{V(\pi)} \pi - J \right) f(\pi) d\pi$$

Proposition 1

The seed investor
maximizes its utility

$\Leftrightarrow$

Uniform valuation
 $V^* = \frac{J}{1-\rho_a-\rho_p^*}$
(binding the cap table)
The seed finances all
projects above this
valuation

What the round leaves

The entrepreneur keeps $\rho_E^* = c_E/\pi_{\min}$ and the seed investor takes $(J - c_E)/\pi_{\min}$: the round buys the company net of the price of running it.

Game 1 (Benchmark)

The pre-seed investor maximizes its utility:

$$\max_{V_p} \underbrace{\int_{\pi_{\min}}^1 \left(\frac{\sigma_p}{V_p} \pi - \sigma_p \right) f(\pi) d\pi}_{\text{projects which will reach seed}} - \underbrace{\int_0^{\pi_{\min}} \sigma_p f(\pi) d\pi}_{\text{projects which will not reach seed}}$$

Proposition 2

The pre-seed investor
maximizes its utility



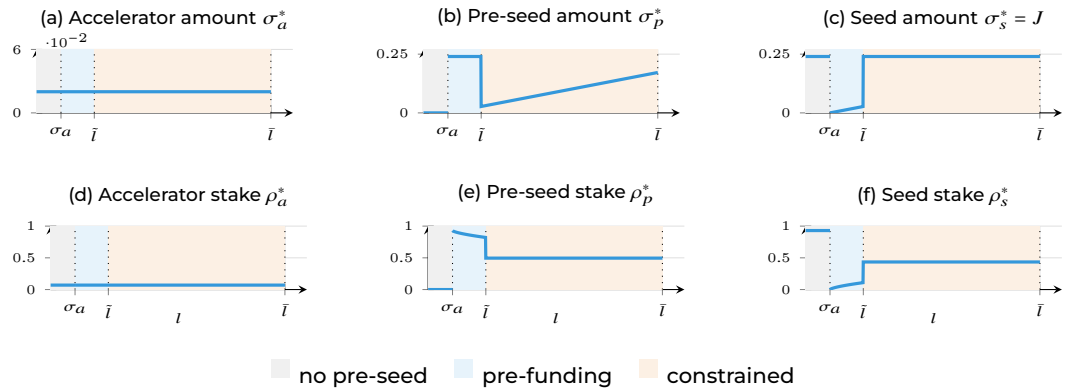
There exists a unique threshold for the pre-seed budget such that:

- **below** only the liquidity shock is covered: $\sigma_p^* = b$
- **above** the budget is saturated, the seed round is pre-funded: $\sigma_p^* = \beta$

The price V_p^* solves the FOC.

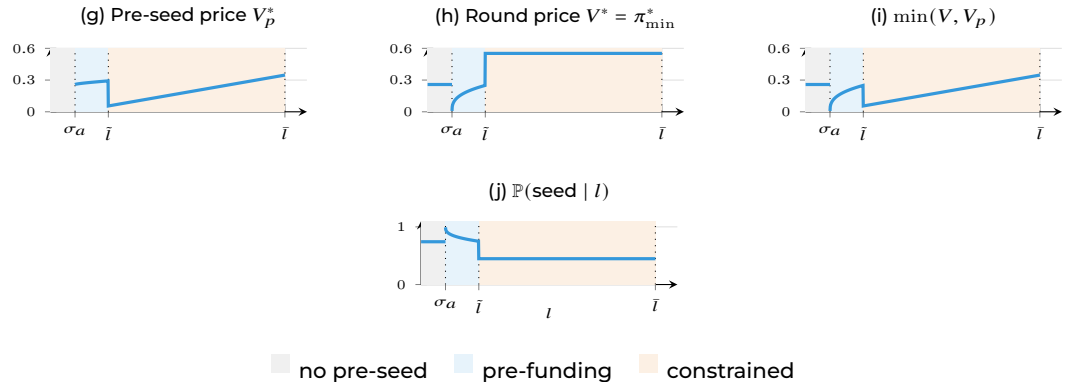
Game 1 (Benchmark)

Below σ_a the pre-seed round is skipped. Between σ_a and $\tilde{l}$ the pre-seed saturates its budget and the seed is left with almost nothing; past $\tilde{l}$ it covers the shock alone and the seed funds J again.



Game 1 (Benchmark)

While it pre-funds, the pre-seed investor prices *above* the round; once constrained, *below* it. The pre-seed investors who pre-fund the seed round increase the seed probability.



Game 2 (MFN)

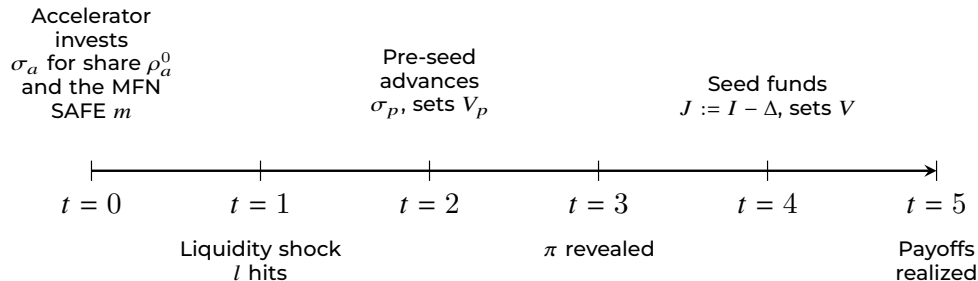


Figure: Game 2 (MFN)

Game 2 (MFN)

V^* and σ_p^* are exactly as in the benchmark.

Proposition 3

There exists a threshold $\bar{\sigma}_p^m$ depending on f, J, m, ρ_a^0 such that the pre-seed investor participates if and only if $\sigma_p^* \leq \bar{\sigma}_p^m$

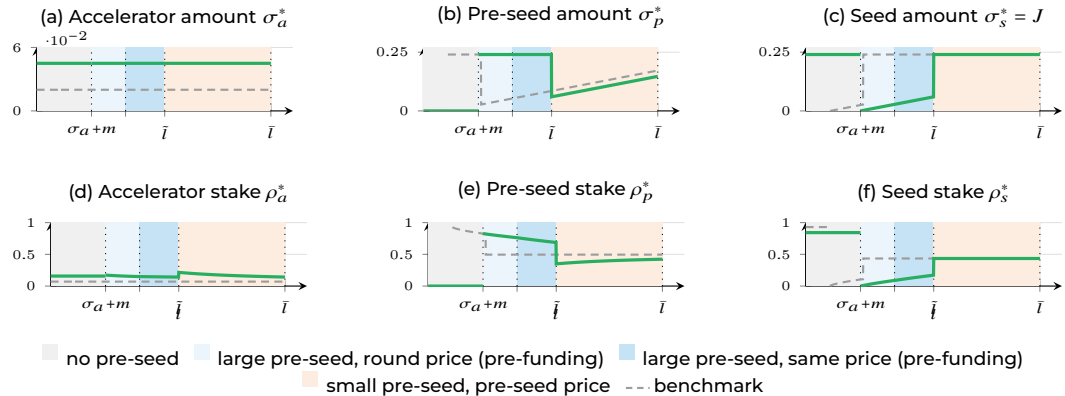
Proposition 4

There exist two thresholds $T_1 < T_2$ depending on f, J, m, ρ_a^0 such that:

- **Small pre-seed, pre-seed price:** $\sigma_p^* \leq T_1$, and $V_p^* < V^*$
- **Large pre-seed, same price:** $T_1 < \sigma_p^* \leq T_2$, and $V_p^* = V^*$
- **Large pre-seed, round price:** $\sigma_p^* > T_2$, and $V_p^* > V^*$

Game 2 (MFN)

Where the investor pre-funds, σ_p is its budget, not the shock: panel (b) is flat across both large bands, and what separates them is T_2 rising past a fixed amount. Elsewhere ρ_a rises by m/V_p^* and ρ_p falls by the same.

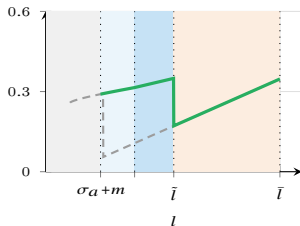


Game 2 (MFN)

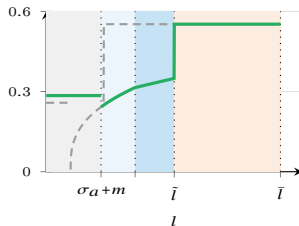
“Remove the pressure to fundraise on unfavourable terms.”

Yes. Fewer firms need a pre-seed round at all, and those that do are priced no lower.

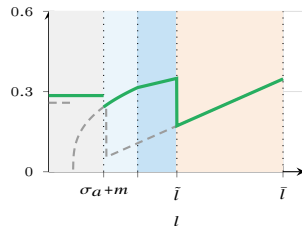
(g) Pre-seed price V_p^*



(h) Round price $V^* = \pi_{\min}^*$



(i) Conversion price $\min(V, V_p)$

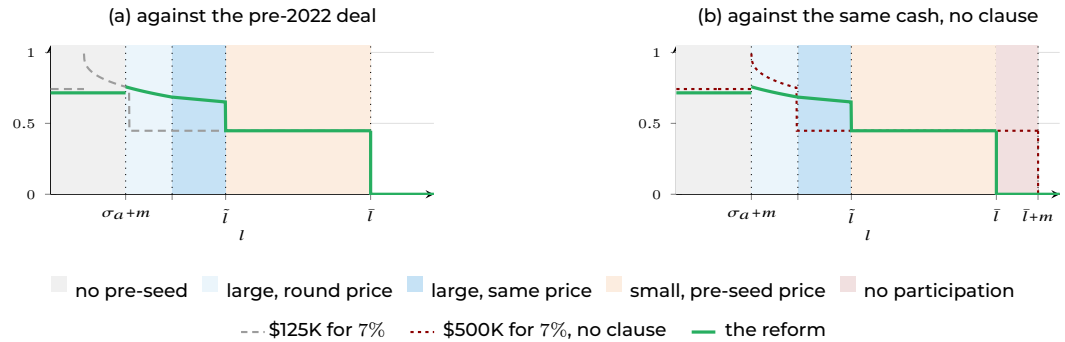


no pre-seed
 large pre-seed, round price (pre-funding)
 large pre-seed, same price (pre-funding)
 small pre-seed, pre-seed price
 benchmark

Game 2 (MFN)

“Make a successful seed round more likely.”

Ambiguous. Against the pre-2022 deal, lower where the accelerator’s cash covers the shock on its own and weakly higher wherever a pre-seed round still occurs. Against the same cash without the clause it moves both ways.



Game 2 (MFN)

Proposition 5

In all three regimes, $\Pi_a^{\text{MFN}} \geq \Pi_a^{\text{bench}}$, strict at every $\sigma_p^* < \bar{\sigma}_p^m$.

Proposition 6

In all three regimes, $\Pi_p^{\text{MFN}} \leq \Pi_p^{\text{bench}}$, with equality only in the large pre-seed, round price regime, at a budget that binds in both games.

Game 2 (MFN)

The seed investor funds less and clears a lower threshold. Neither effect dominates.

Proposition 7

$$\Pi_s^{\text{MFN}} - \Pi_s^{\text{bench}} = \underbrace{(J^{\text{MFN}} - J^{\text{bench}}) S(V^*)}_{\leq 0} + \underbrace{J^{\text{bench}} [S(V^*) - S(\pi^{\text{bench}})]}_{\geq 0},$$

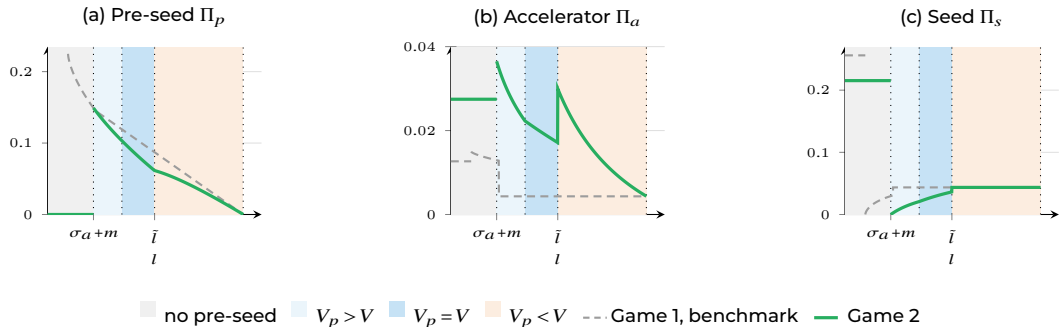
writing $\Pi_s = J S(V)$, $S(V) = M(V)/V - (1 - F(V))$ decreasing. Both terms vanish where the investor advances the shock and no more; the second in the large pre-seed, round price regime at a budget that binds in both games, leaving $-m S(V^*)$.

Corollary

There the seed investor is untouched, and what the accelerator gains the pre-seed investor loses: $\Pi_a^{\text{MFN}} - \Pi_a^{\text{bench}} = -(\Pi_p^{\text{MFN}} - \Pi_p^{\text{bench}}) = \frac{m}{\sigma_p^*} \Pi_p^{\text{bench}}$.

Game 2 (MFN)

The accelerator earns more at every shock, up to +0.030. The pre-seed investor earns less, down to -0.026. The seed investor is untouched past $\tilde{l}$, earns up to 0.041 less before it, and more on the grey band, where the extra cash makes a pre-seed round unnecessary.



Game 3 (capped SAFE)

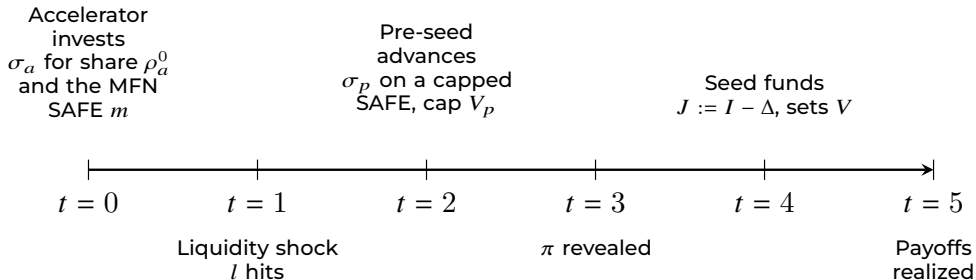


Figure: Game 3 (capped SAFE)

Game 3 (capped SAFE)

Both claims now convert at $\min(V, V_p)$, so $\rho_p = \sigma_p / \min(V, V_p)$.

Proposition 8

T_2 disappears: only one threshold is left, and two regimes instead of three:

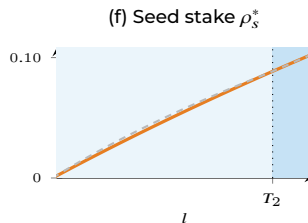
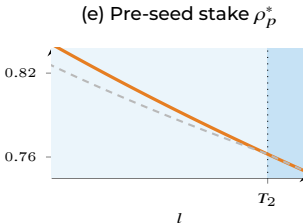
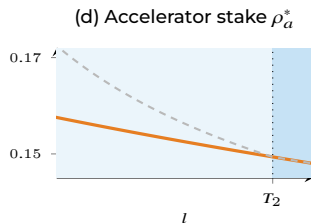
- $\sigma_p^* \leq T_1$: $V_p^* < V^*$, both claims convert at the pre-seed price, threshold π_1
- $\sigma_p^* > T_1$: V_p^* indeterminate, both claims convert at the round price, threshold $\hat{V}_p$

Proposition 9

The pre-seed investor's payoff is never higher under the cap than under a price it sets itself, and strictly lower when $\sigma_p^* > T_2$ — an amount it advances only if $\pi_1 < \bar{x}$.

Game 3 (capped SAFE)

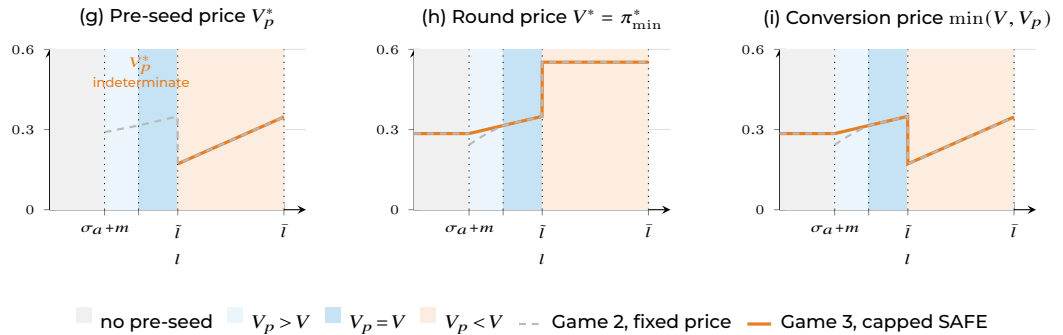
The amounts are identical at every shock, so only the stakes are drawn, and only on the window where the two games differ at all — the band where Game 2's investor prices above the round. Even there, nothing moves by more than 1.5 points.



no pre-seed $V_p > V$ $V_p = V$ $V_p < V$ -- Game 2, fixed price — Game 3, capped SAFE

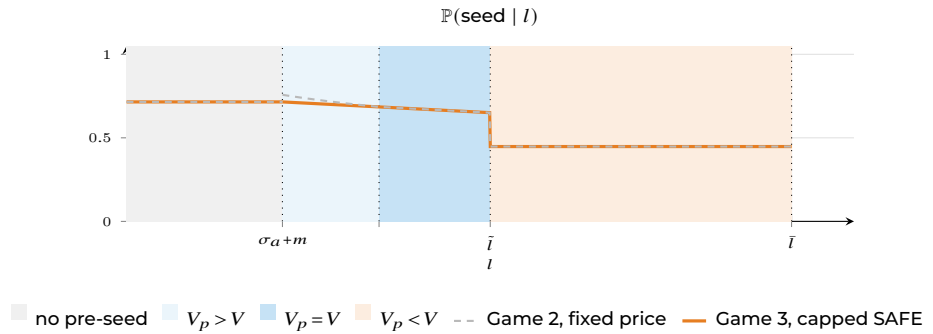
Game 3 (capped SAFE)

Where the cap does not bind, V_p^* is indeterminate: every base above $\widehat{V}_p$ pays the same, so (g) carries no Game 3 curve. The round then prices at $\widehat{V}_p$, above the π_2 Game 2 attains.



A cap that never binds is a price

The cap costs its holder nothing where it binds — there it is the price the investor would have named anyway — and up to four points of $\mathbb{P}(\text{seed} \mid l)$ where it does not.



Game 3 (capped SAFE)

The two games differ only on $\sigma_p^* > T_2$: the threshold is π_2 when the pre-seed investor sets its own price, and $\widehat{V}_p > \pi_2$ under the cap.

Proposition 10

At a given amount,

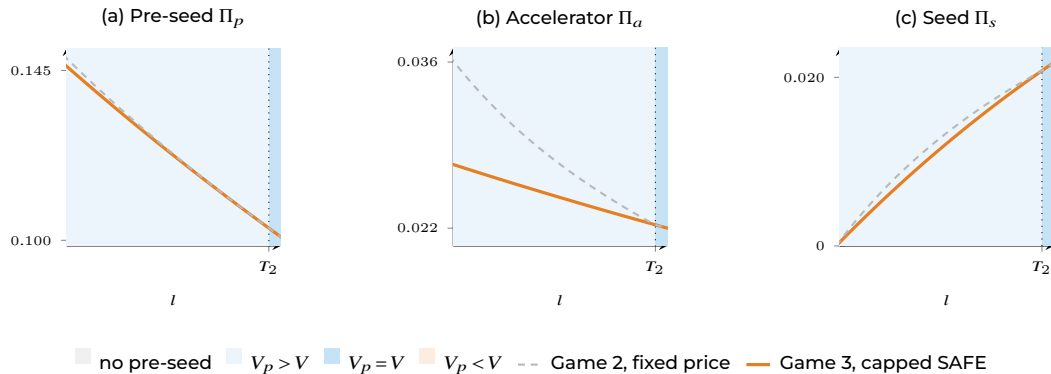
$$\Pi_p^{G3} \leq \Pi_p^{G2}, \quad \Pi_a^{G3} \leq \Pi_a^{G2}, \quad \Pi_s^{G3} \leq \Pi_s^{G2},$$

with equality if and only if $\sigma_p^* \leq T_2$, and all three strict beyond it.

Each payoff is strictly decreasing in the threshold above π_2 : $\Pi_p = \varphi_{J+m}(V)$, which π_2 maximizes; $\Pi_a = \rho_a^0 M(V) + m M(V)/V - \sigma_a - m$; and $\Pi_s = J S(V)$. The table is saturated in both games, so nobody takes what the three give up.

Game 3 (capped SAFE)

Outside this window the two games coincide exactly.
Inside it the cap costs its own holder 1.5% of its payoff at worst, the accelerator 25% and the seed investor 24% — all three closing at T_2 , where the cap stops binding.



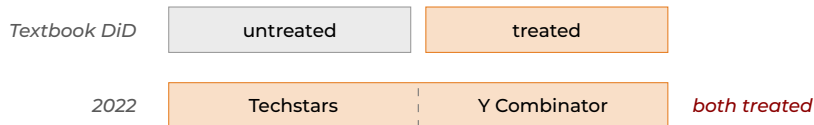
Crunchbase, Y Combinator and Techstars cohorts 2019-H1 to 2021-H2: 962 and 715 firms.

- Outcomes read over a fixed 24-month window from the batch date
- σ_a measured batch by batch, not assumed: \$150K, then \$125K, then \$500K
- The reform lands on the January 2022 batch — 97% of its rounds at the new amount

What Crunchbase does not see: valuations. One Y Combinator firm reports both the interim and the priced valuation — and one at Techstars.

The problem: the control is treated too

A market-wide correction hits both accelerators in 2022. The textbook contrast does not exist.



Techstars changed nothing and lost 13.1 pp of interim rounds. Identification comes from the model, not from a control group.

What the moments identify

Eleven moments, all on the pre-reform cohorts, where the clause is absent.

	Moment	Measured on	Identifies
$M1$	Pr(interim round)	all 962	the window of G , and p_0
$M2$	Interim amounts, 3 quantiles	174	the scale of G
$M3$	Pr(priced), both arms	all 962	F and the threshold
$M4$	Priced amounts, 3 quantiles	106	I , hence c_E
$M5$	Interim over priced	26	the corner: Δ

Simulated method of moments; standard errors are firm-level bootstrap, 200 draws.

Only the entrepreneur's cost makes the model fit

		Criterion	d.f.	p	Verdict
A	$c_E = 0$	26.0	4	$2 \cdot 10^{-6}$	rejected
B	$c_E = 0$, budget, entry cost	42.7	2	$< 10^{-3}$	rejected
C	c_E free	3.40	3	0.15	not rejected
D	c_E free, budget, entry cost	4.86	1	0.04	rejected at 5%

Setting $c_E = 0$ asks the investment cost alone to span interim and priced rounds. It cannot. $\hat{c}_E = 0.20$, $\hat{\rho}_E = 0.56$, median shock \$4.7M.

These firms bridge, they do not pre-fund

Which regime

$$\frac{\sigma_p}{J} = 0.21 \quad \Rightarrow \quad \Delta \approx 0 \quad \Rightarrow \quad \sigma_p \leq W(J) - m \leq T_1$$

No participating investor reaches T_1 , in all 200 draws. The clause stays in the small pre-seed, pre-seed price regime: it redistributes, and the threshold is the benchmark's.

The reform moves graduation by a tenth of a point

Percentage points	Cash	Clause	Total	95% interval
Pr(priced round)	+0.31	-0.45	-0.09	[-0.16, +0.02]
Pr(interim round)	+1.59	-1.93	-0.35	[-0.88, -0.10]

The cash enters through the threshold, the clause through participation. Graduation actually fell 13.8 points — which would take m at 90 times its size.

What reproduces the post-reform cohorts

Quality and financing needs, moved together. The contract alone does not.

	Freed, post-reform contract imposed	Criterion	d.f.	p	Verdict
(0)	Nothing	1091.9	9	$< 10^{-3}$	rejected
(a)	F , quality	184.6	7	$< 10^{-3}$	rejected
(c)	I , cost of the priced round	43.4	7	$3 \cdot 10^{-5}$	rejected
(b)	G , liquidity shocks	14.4	6	0.033	rejected
(d)	F and G	5.68	4	0.30	not rejected

$\mathbb{E}(\pi) -25\%$, $\sigma(\pi) +22\%$; median shock $+151\%$. Neither is attributable to the change of terms.

The reduced-form design fails its own test

We ran it. Y Combinator against Techstars, cohorts around January 2022.

Outcome	Joint pre-trend Wald	Verdict
Pr(priced), 24m	635	pre-trends fail
Pr(priced), 36m	211	pre-trends fail
log interim amount	16.3	pre-trends fail

Graduation falls $0.282 \rightarrow 0.126$ before the clause exists — half the total decline, inside the control period. The estimand is 0.09 pp; 2,000 firms resolve about 1.

What we cannot say

- **Which regime, from prices.** The price ratio sorts the three regimes, and one firm in the panel reports both valuations. The corner is read off amounts instead.
- **Whether interim rounds grew.** The model's one unambiguous prediction. Coverage of interim amounts falls from 58% to 36% at Y Combinator while rising at Techstars.
- **That the reform caused anything.** The counterfactual is the model run twice. It bounds the clause; it does not identify its effect.

Main result

The accelerator is the only winner of the MFN introduction. The pre-seed investor strictly loses. The probability of seed success and the seed investor's gain are both ambiguous — and estimated on Y Combinator, the reform moves graduation by -0.09 points.

- **The liquidity shock decides where the tranche converts.** A small one leaves it in the pre-seed block; a large one carries it into the round.
- **Agent by agent.** The accelerator earns more at every shock and the pre-seed investor less. The seed investor is untouched where that investor advances the shock and no more, and its payoff has no sign where it pre-funds.
- **A capped SAFE is worse for the pre-seed investor.** Above T_1 the cap stops binding, the claim converts at V , and every payoff falls.
- **The clause costs financing states even where its price is absorbed.** The cap on the amount falls to $W(J) - m$, so companies in that band lose their investor. Only the accelerator's cash pushes the other way.

Thank you
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